Taste Heterogeneity and the Linear Context Logit: Re-examining the Sushi Context Effect

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***Abstract*—The linear context logit (LCL) of Tomlinson and Benson lets the mean features of a choice set change a chooser's preferences, and tests each such context effect with a likelihood-ratio test. On the sushi data, where menus are random, the paper finds that people avoid oily sushi more when the menu is oilier ($A_{\text{oil,oil}} = -0.27$, $p = 1.6 \times 10^{-6}$) and suggests that the effect may be causal. We test whether the same pattern can arise because people differ in how much they like oily food. A second-order expansion shows that a population with taste variance s^2 behaves like a single chooser whose oiliness weight falls by s^2 times the menu's weighted mean oiliness, which an LCL reads as a negative self-effect. In simulations on the real sushi menus with no context effect, a normal spread of tastes or two segments of people reproduce the paper's estimate and its significance for $s \approx 0.8$–1.0. The same test run inside a random-coefficient logit flags 0–1% of these datasets and still detects a true effect of the paper's size in 91% of cases. On the real data the fitted taste spread is 0.45. A parametric bootstrap attributes about a third of the effect to it, and inside heterogeneity models the effect falls to -0.21 with $p \approx 10^{-3}$. Using each respondent's second and third ranked choices, the effect is -0.06 ($p = 0.23$) and $+0.02$ ($p = 0.73$), with a test that we show detects the paper's effect size in 94–100% of simulated datasets. A first-pick effect of about -0.15 ($p \approx 0.02$) remains. The paper's decomposed model (DLCL), which already contains latent classes, keeps its false-positive rate under heterogeneity but has little power for this effect, and it does not support the oiliness effect either once repeated choices are used.**

***Index Terms*—Discrete choice, context effects, linear context logit, mixed logit, preference heterogeneity.**

I. Introduction

MOST discrete choice models assume independence of irrelevant alternatives (IIA): the relative odds of two items do not depend on what else is offered. Behavioural work finds many violations, called context effects. Tomlinson and Benson [1] propose the linear context logit (LCL), in which the mean features of the choice set shift the chooser's preference vector through a matrix A. The model keeps a concave log-likelihood, and each entry A_{pq} has a direct reading (how feature q on the menu changes the weight on feature p) and a likelihood-ratio test (LRT).

The paper's main interpretable result is on the sushi data [2]. Each of 5,000 respondents ranks a random set of 10 sushi drawn from 100. The most significant effect is the oiliness self-effect, $A_{\text{oil,oil}} = -0.20$ in the full model and -0.27 when fitted alone (LRT 23, $p = 1.5 \times 10^{-6}$). Because the menus are random, the authors suggest that it may be a causal effect, in which an oily menu makes oily sushi less appealing.

Random menus remove choice-set confounding [3], where menus depend on the chooser. They do not remove taste heterogeneity, where different people have different coefficients. If some people like oily sushi and others avoid it, an oily menu spreads the first group's choices over many oily items while the second group concentrates on the few light ones. Each oily item is then chosen less often on an oily menu, although no individual's preferences depend on the menu. The paper compares LCL with a mixed logit as a separate model, but it does not test the oiliness effect inside a model that allows tastes to vary.

Our question is whether LCL can tell a context effect apart from taste heterogeneity, and whether the sushi oiliness effect remains once heterogeneity is modelled.

We think this is worth asking for two reasons. First, what makes LCL attractive is that each entry of A can be read and tested on its own. If heterogeneity alone produces significant entries, then a significant entry tells us that the data are not MNL, but not that the menu changes anyone's preferences, and causal readings like the one above need a different test. Second, the question can actually be answered. A context effect acts on every choice a person makes from a menu, while a taste difference stays with the person. So the two explanations make different predictions as soon as we observe the same person choosing more than once, and the sushi data, which record a full ranking for every respondent, let us do that.

Our approach has four parts. We first reproduce the paper's numbers. We then work out what heterogeneity

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looks like to an LCL, using a second-order expansion. Next, we simulate choices on the real sushi menus from ground truths that we control, including two kinds of heterogeneity without any context effect, and compare the paper's test with the same test run inside a heterogeneity model. Finally, we return to the real data.

Our contributions are:

- An exact reproduction of the paper's sushi results.
- A second-order expansion showing why taste heterogeneity appears to an LCL as a negative self-effect, checked numerically on the real menus.
- A simulation study on the real sushi menus with two kinds of heterogeneity (a continuous normal spread and discrete segments), comparing the paper's test with the same test inside heterogeneity models.
- A re-analysis of the real data: tests inside heterogeneity models, a parametric bootstrap, and a panel analysis that uses the full rankings the paper discards, with a simulation check of that test.

A. Related Work

That averaging over people can create context effects is known outside machine learning. Hutchinson, Kamakura and Lynch [4] showed that context ("reversal") effects in choice experiments can arise from pooling segments of consumers that do not show the effect separately. Liew, Howe and Little [5] analysed participants in classic context-effect experiments individually and found that few showed the effects seen in the averaged data. In econometrics, the distinction between true and spurious state dependence [6] has the same structure, and is resolved with repeated observations of the same person. Mixed logit can approximate any random utility model [7], and the similarity effect of Tversky [8] is the standard example of a pattern it produces without context effects [9]. Marketing models include explicit context-effect terms next to context-free part-worths so that both are estimated together [10].

In machine learning, LCL follows the context-dependent random utility model of Seshadri et al. [11]. Seshadri and Ugander [12] show that testing IIA in general needs a number of samples exponential in the number of alternatives. Tomlinson, Ugander and Benson [3] show that choice sets that depend on the chooser bias preference estimates. The LCL paper itself does compare a model with context effects and latent classes (the DLCL) against a latent-class mixed logit, and on sushi reports LRT 59, $p = 0.0089$. We find that both models were fitted well short of their optimum, and that with better fits this comparison is highly significant (Section III-H). That test covers all context effects at once, however; the oiliness effect itself is tested only against a plain MNL. We did not find work that tests LCL's individual effects inside a heterogeneity model.

II. Methods

A. Models

A chooser picks item i from a set C, and item i has features $x_i \in \mathbb{R}^d$. The multinomial logit (MNL) has $\Pr(i \mid C) \propto \exp(\theta^\top x_i)$. The LCL [1] uses a preference vector that depends on the set:

$$\Pr(i \mid C) \propto \exp\left((\theta + A\bar{x}_C)^\top x_i\right), \qquad \bar{x}_C = \frac{1}{|C|}\sum_{j \in C} x_j. \tag{1}$$

An entry A_{pq} is tested with an LRT against MNL in which only that entry is free (the paper's constrained LCL). We use two models of taste heterogeneity in which every individual is an MNL chooser [9]:

- random coefficients (RC): $\theta_n = \theta + \eta_n$, $\eta_n \sim \mathcal{N}(0, \mathrm{diag}(\sigma^2))$ on selected features;
- latent classes (LC): $\theta_n = \theta + \delta_c$ with probability π_c, $c = 1, \ldots, K$.

Both can be combined with A. The test of $A_{\text{oil,oil}}$ inside RC or LC is a one-degree-of-freedom LRT between the heterogeneity model with and without that entry. When a person makes several choices, η_n is shared across them (a panel model).

The paper's second model, the decomposed LCL (DLCL), is a mixture of d logit components. Component q has its own preference vector B_q and the context effect of feature q only, so the coefficient of feature p in component q is $B_{qp} + A_{pq}\bar{x}_{C,q}$. With $A = 0$ the DLCL is a latent-class logit with d classes, which is the paper's "mixed logit" (ML6 below, since $d = 6$). The DLCL therefore already contains a model of discrete taste heterogeneity, and its test against ML6 is a test of context effects that allows for it.

B. Heterogeneity Seen Through an LCL

To see why heterogeneity can look like a context effect, consider the simplest case. Person n has $\theta_n = \mu + \eta_n e_k$, so that everyone agrees except on feature k, with $\mathbb{E}\eta_n = 0$ and $\mathrm{Var}\,\eta_n = s^2$. For a set C, let p_j be the MNL probabilities at the average taste μ, and let $m_C = \sum_j p_j x_{jk}$ and $v_C = \sum_j p_j (x_{jk} - m_C)^2$ be the weighted mean and variance of feature k in the set.

Proposition 1: The population choice probabilities $P_i = \mathbb{E}_\eta \Pr_{\mu+\eta e_k}(i \mid C)$ satisfy

$$\log P_i = \mu^\top x_i + \frac{s^2}{2}x_{ik}^2 - s^2 m_C\, x_{ik} - Z_C + O(\mathbb{E}|\eta|^3), \tag{2}$$

where Z_C is the same for every item in C.

Proof: Let $f(\eta)$ be the log-probability of item i at taste $\mu + \eta e_k$. Differentiating the log-softmax gives $f'(0) = x_{ik} - m_C$ and $f''(0) = -v_C$, so $P_i = p_i\big(1 + \frac{s^2}{2}[(x_{ik} - m_C)^2 - v_C]\big) + O(\mathbb{E}|\eta|^3)$. Taking logs and expanding the square, the terms m_C^2 and v_C do not depend on i and go into Z_C. ∎

In words, a population of ordinary logit choosers who disagree about feature k behaves, to second order, like one chooser with two extra terms. The first is a curvature term in x_{ik}. The second is a coefficient on feature k that falls as the menu's weighted mean of k rises, which is exactly the shape of a negative LCL self-effect. In sushi, a regression of m_C on the plain set mean $\bar{x}_{C,\text{oil}}$ has slope $\beta = 1.16$, so an LCL fitted to such data should report $A_{kk} \approx -\beta s^2$. Two further points follow. The LCL has no x_{ik}^2 term, so part of the curvature is absorbed elsewhere and the fitted value is smaller in size; adding a squared term removes this attenuation and makes the spurious effect larger rather than removing it. And since only the variance s^2 enters at this order, a normal spread and a population of segments should give the same effect when s is small.

There is also a limit to what heterogeneity can do. Any mixture of logits is a random utility model, and every random utility model satisfies regularity: adding an item to a menu never raises the probability of an item that was already there [13]. Heterogeneity can therefore produce negative self-effects of the similarity type, but never an effect that breaks regularity, such as the attraction effect [14].

C. Pipeline of the Simulation Study

Fig. 1 shows the simulation design. We keep the 5,000 real sushi menus and their item features, and simulate only the choices, from a ground truth that we choose. The base tastes are the MNL fit to sushi. The ground truths are: no effect (MNL); a true effect $A_{\text{oil,oil}} = -0.266$ with identical people; a normal spread of the oiliness coefficient with standard deviation s; two segments with equal shares, or with shares 20/80, whose oiliness coefficients have variance s^2; and a smaller true effect (-0.15) together with a normal spread ($s = 0.75$). Each simulated dataset is fitted with MNL, the full LCL, the one-entry LCL, and three models with and without $A_{\text{oil,oil}}$: a random coefficient on oiliness (RC), two latent classes on oiliness (LC2), and an MNL with a squared oiliness term (SQ). We use the paper's threshold $p < 0.001$ and 100 replicates per setting.

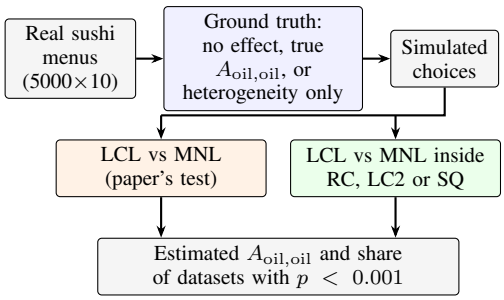


Fig. 1. Simulation design. Menus and item features are the real sushi ones; only the choices are simulated.

D. Dataset

The sushi data [2] has 5,000 respondents, each ranking 10 of 100 sushi from most to least preferred. Following the authors' code, we take the top-ranked item as the choice and use six features, standardised over all item occurrences: is maki, is seafood, oiliness, popularity, price and availability. Oiliness is coded from 0 (oily) to 4 (light); the self-effect $A_{\text{oil,oil}}$ does not depend on the direction of the coding. The data also include gender, age group and whether the respondent grew up in eastern or western Japan.

For the panel analysis we treat a ranking as successive choices (the rank-ordered logit [15]): the best of 10, then the best of the remaining 9, and so on. With the top k ranks each respondent gives k choices, made from menus whose mean features change from stage to stage.

E. Implementation Details

All models are written in PyTorch as one class with optional A entries, squared terms, demographic interactions, random coefficients and latent classes. We fit by exact maximum likelihood with L-BFGS in double precision and without regularisation, so that the LRTs are not affected by a penalty. RC likelihoods use 200 scrambled Sobol draws per person (100 in panel models). Latent-class models use several random restarts. Nested models are fitted starting from each other's optimum, which keeps every LRT statistic non-negative. We only test entries of A; LRTs for variance parameters lie on the boundary of the parameter space and are not used. The simulation study took about four hours on 8 CPU cores, and the bootstrap and panel checks about one hour together.

III. Results

A. Reproduction

Table I compares our fits with the paper. The numbers agree to the reported precision. Two other single effects

TABLE I
REPRODUCTION OF THE PAPER'S SUSHI RESULTS (LRT: ALL 36 ENTRIES OF A)

	MNL NLL	LCL NLL	LRT	$A_{\text{oil,oil}}$ full / alone
Paper [1]	9821	9773	97	-0.20 / -0.26
Ours	9821.4	9772.7	97.3	-0.204 / -0.266

TABLE II
FITTED COEFFICIENTS OF THE TWO TERMS IN (2), OVER THREE TASTE DISTRIBUTIONS WITH VARIANCE s^2

	x_{ik}^2		$m_C\, x_{ik}$	
s	theory	fitted	theory	fitted
0.25	0.031	0.022 to 0.031	-0.062	-0.060 to -0.051
0.50	0.125	0.100 to 0.121	-0.250	-0.203 to -0.183
1.00	0.500	0.250 to 0.318	-1.000	-0.511 to -0.385

are significant at $p < 0.001$: price on price (-0.18) and price on oiliness ($+0.18$).

B. Check of the Expansion

We simulated 100,000 choices on the real menus (each menu used 20 times) with heterogeneity in oiliness only, and fitted an MNL with the two regressors of (2), x_{ik}^2 and $m_C x_{ik}$, added (Table II). For $s = 0.25$ the fitted coefficients are close to $s^2/2$ and $-s^2$ for all three taste distributions. For larger s the signs stay correct but the sizes are smaller than predicted, and the three distributions start to differ, which is the expected effect of third-order terms.

C. Simulation Study

Fig. 2 shows the mean $A_{\text{oil,oil}}$ reported by the one-entry LCL when no one in the population has a context effect. The estimate grows with the taste spread and reaches the paper's -0.27 at $s \approx 0.8$–1.0. This is the central result of the simulation: the effect the paper reports is what LCL finds in a population of context-free choosers whose tastes differ by about one standard deviation. The three taste distributions agree at $s = 0.5$, as the proposition predicts, and segments give a larger effect than a normal spread with the same variance at larger s. In our first pilot, which used synthetic Gaussian item features instead of the sushi items, the same spread gave a much smaller effect (-0.06 at $s = 1$), so the size of the bias depends on the menu design.

Fig. 3 and Table III give the rejection rates. The paper's test flags an effect in 57–78% of datasets at $s = 0.75$ and in 92–99% at $s = 1$. The same test inside RC or LC2 flags 0–1% of these datasets at every

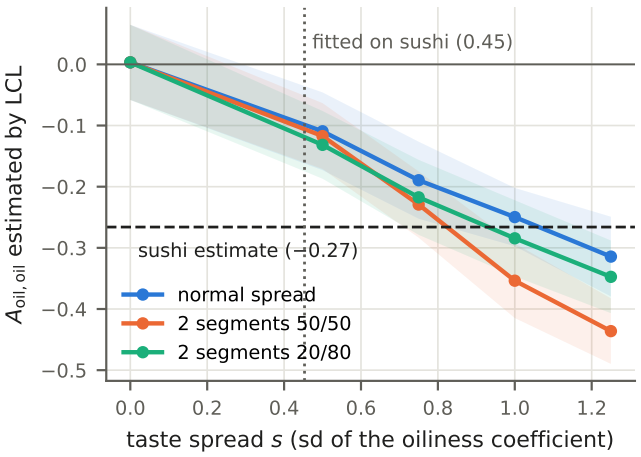


Fig. 2. Mean $A_{\text{oil,oil}}$ estimated by the one-entry LCL on simulated data with no context effect, against the spread s of the oiliness coefficient (shading: ± 1 sd over 100 replicates). The dotted line is the spread fitted on the real data.

s, and detects the true effect in 91% (RC) and 88% (LC2) of datasets, against 94% for the paper's test. So modelling heterogeneity removes the false positives while costing very little power, and this holds whether the heterogeneity is continuous or comes from segments. When a true effect of -0.15 is combined with heterogeneity, RC estimates -0.14 while the paper's test reports -0.30; at this smaller effect size RC detects it in only 10% of datasets. A normal RC fitted to data from two segments leaves a small negative estimate at large s (-0.08 to -0.11) but rarely a significant one. The squared term does not help: inside SQ the false-positive rate is 45–100%, as the expansion predicts. Finally, at equal parameter counts, RC fits better than the one-entry LCL in all heterogeneity datasets and in none of the true-effect datasets.

D. Heterogeneity Models on the Sushi Data

Table IV tests $A_{\text{oil,oil}}$ inside several heterogeneity models fitted to the paper's data. Letting the oiliness coefficient vary adds one parameter and improves the log-likelihood by 13.9 over MNL. It also fits better than the one-entry LCL (-9809.8), which has the same number of parameters. Inside every heterogeneity model the effect falls to about -0.21 and its p-value rises to about 10^{-3}, the paper's threshold. Demographics change tastes (women prefer lighter sushi) but do not change $A_{\text{oil,oil}}$. Inside RC on all features, price on price goes from $p = 4 \times 10^{-4}$ to 0.014 and price on oiliness from 6×10^{-4} to 0.039. The test of all 36 entries of A remains significant (LRT 97 to 78, $p = 7 \times 10^{-5}$).

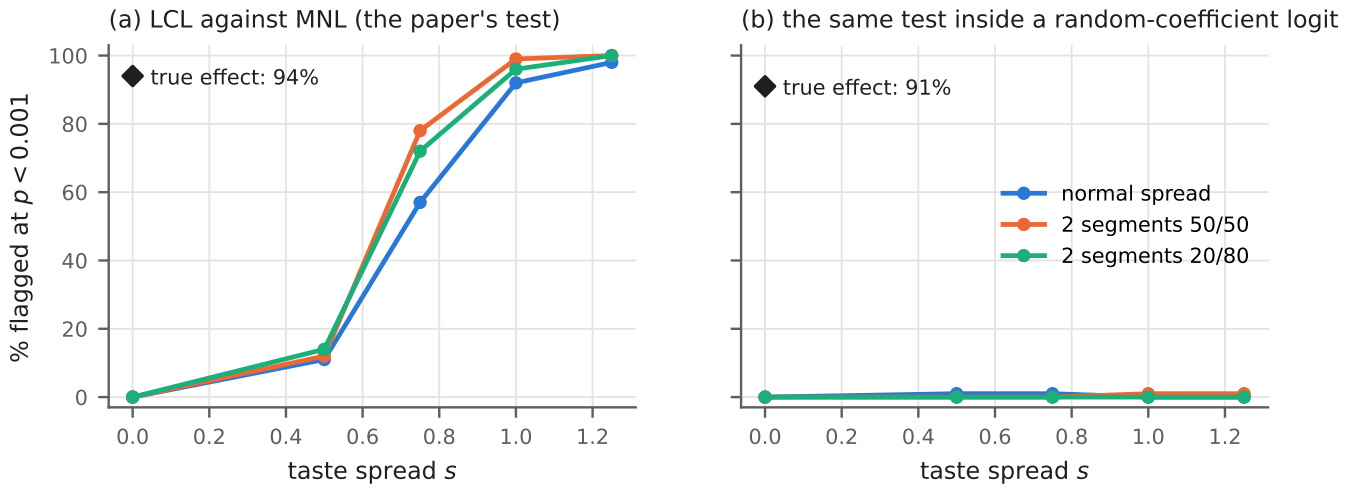


Fig. 3. Share of simulated datasets in which $A_{\text{oil,oil}}$ is significant at $p < 0.001$. (a) The paper's test. (b) The same test inside a random-coefficient logit. Diamonds: datasets with a true effect of the paper's size and no heterogeneity.

TABLE III
SIMULATION RESULTS (100 REPLICATES PER ROW; RATES VARY BY ABOUT ±5–10 POINTS). *SIGNIFICANT AND AT LEAST AS NEGATIVE AS THE PAPER'S −0.266

		LCL against MNL				test of $A_{\text{oil,oil}}$ inside …			
Ground truth	s	mean $A_{\text{oil,oil}}$	36-df	single	as paper*	RC: mean $A_{\text{oil,oil}}$	RC	LC2	SQ
no effect	0	+0.00	0%	0%	0%	+0.01	0%	0%	0%
true effect −0.27	0	−0.26	21%	94%	44%	−0.25	91%	88%	91%
effect −0.15 + normal spread	0.75	−0.30	75%	98%	72%	−0.14	10%	6%	100%
normal spread	0.50	−0.11	2%	11%	1%	−0.00	1%	0%	45%
	0.75	−0.19	33%	57%	11%	−0.00	1%	1%	98%
	1.00	−0.25	88%	92%	39%	+0.02	0%	0%	100%
	1.25	−0.31	100%	98%	77%	+0.01	0%	0%	100%
two segments, 50/50	0.50	−0.12	3%	12%	0%	−0.00	0%	0%	57%
	0.75	−0.23	61%	78%	25%	−0.02	0%	0%	100%
	1.00	−0.35	100%	99%	94%	−0.08	1%	0%	100%
	1.25	−0.44	100%	100%	100%	−0.11	1%	0%	100%
two segments, 20/80	0.50	−0.13	4%	14%	2%	−0.01	0%	0%	76%
	0.75	−0.22	60%	72%	26%	+0.00	0%	0%	100%
	1.00	−0.28	99%	96%	60%	+0.04	0%	0%	100%
	1.25	−0.35	100%	100%	93%	+0.05	0%	0%	100%

E. Parametric Bootstrap

We fitted heterogeneity models without context effects to the sushi data, simulated 200 datasets from each on the real menus, and repeated the paper's analysis (Fig. 4). With the fitted spread of 0.45, the mean estimate is −0.09, about a third of the observed −0.27. The effect is significant at $p < 0.001$ in 3–6% of these datasets (12% for a three-class model) and as negative as −0.266 in at most 0.5%. None reaches the paper's 36-entry LRT of 97. With one choice per person, the fitted heterogeneity explains part of the effect but not all of it.

F. Panel Analysis with the Full Rankings

With one choice per person, a person's taste and the menu are hard to separate. Using the top k ranks gives k choices per person with a shared taste η_n and a menu mean that changes between stages. Table V shows the result. Inside a panel RC on all features, the effect is −0.06 ($p = 0.23$) with $k = 2$ and +0.02 ($p = 0.73$) with $k = 3$, while an LCL without heterogeneity becomes more significant as k grows.

To check that the panel test could have found the effect, we simulated 50 ranking datasets per ground truth on the real menus, choosing one stage at a time so that

TABLE IV
TEST OF $A_{\text{oil,oil}}$ INSIDE HETEROGENEITY MODELS, SUSHI DATA WITH ONE CHOICE PER PERSON

Base model	log-lik.	$A_{\text{oil,oil}}$	LRT (p)
MNL (paper)	-9821.4	-0.266	23.1 (1.6×10^{-6})
MNL + squared oiliness	-9820.7	-0.296	27.5 (1.5×10^{-7})
RC, oiliness (sd 0.45)	-9807.5	-0.214	10.9 (9.4×10^{-4})
RC, all features	-9804.3	-0.220	10.7 (1.0×10^{-3})
2 classes, oiliness	-9807.1	-0.214	10.9 (9.6×10^{-4})
2 classes, all features	-9799.0	-0.207	9.8 (1.7×10^{-3})
Demographics	-9757.5	-0.265	22.7 (1.9×10^{-6})
Demographics + RC, all	-9741.0	-0.216	10.4 (1.2×10^{-3})

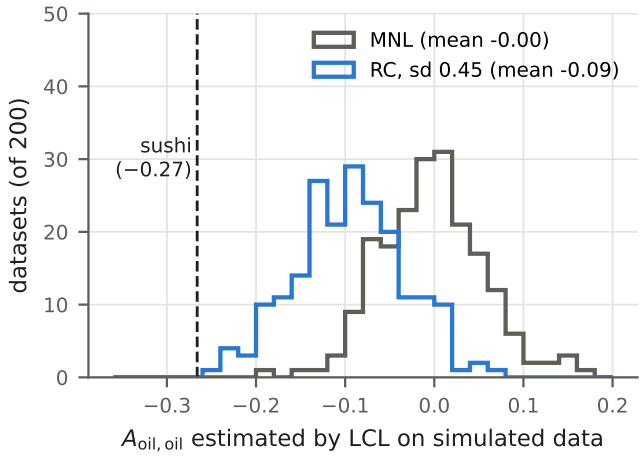


Fig. 4. Parametric bootstrap. $A_{\text{oil,oil}}$ estimated by the one-entry LCL on 200 datasets simulated from MNL and from a random-coefficient logit fitted to sushi. Models with two classes or with all features heterogeneous give almost the same distribution (mean -0.09).

TABLE V
SUSHI DATA WITH THE TOP k RANKED CHOICES PER PERSON

k	LCL: $A_{\text{oil,oil}}$ (p)	panel RC + $A_{\text{oil,oil}}$: $A_{\text{oil,oil}}$ (p)
1	-0.266 (1.6×10^{-6})	-0.220 (1.0×10^{-3})
2	-0.243 (1.6×10^{-10})	-0.064 (0.23)
3	-0.228 (5.7×10^{-14})	$+0.015$ (0.73)

a true LCL effect acts on the remaining items at each stage, with tastes from the sushi panel fit (Table VI). With heterogeneity and no context effect, the panel test flags no dataset, while the LCL without heterogeneity flags 78% ($k = 2$) and 100% ($k = 3$). With a true effect of -0.266, the panel test detects it in 94–100% of datasets. In ranking data the later menus depend on each person's earlier picks, so heterogeneity becomes choice-set confounding [3], which explains the growing bias of the plain LCL.

TABLE VI
SIMULATION CHECK OF THE PANEL TEST (50 DATASETS PER ROW; FLAGGED: $p < 0.001$)

		LCL		panel RC + $A_{\text{oil,oil}}$	
Ground truth	k	$A_{\text{oil,oil}}$	flagged	$A_{\text{oil,oil}}$	flagged
Heterogeneity only	2	-0.16	78%	$+0.00$	0%
	3	-0.22	100%	$+0.01$	0%
Heterogeneity + effect	2	-0.38	100%	-0.25	94%
	3	-0.43	100%	-0.26	100%
Effect only	2	-0.26	100%	-0.26	100%
	3	-0.27	100%	-0.26	100%

G. First Pick and Later Picks

We then allowed separate self-effects for the first pick, made from the full menu, and for later picks. Without heterogeneity both stages show the effect (-0.23 and -0.25 with $k = 2$, both $p < 10^{-4}$). Inside the panel RC, the later picks show no effect ($+0.04$, $p = 0.61$ with $k = 2$; $+0.09$, $p = 0.06$ with $k = 3$), while the first pick keeps one of -0.16 ($p = 0.017$) with $k = 2$ and -0.14 ($p = 0.034$) with $k = 3$. This could be a reaction to the full menu at the first choice, or heterogeneity that a diagonal normal model does not capture, such as correlated tastes. Our data do not separate the two.

H. The Decomposed LCL

Because the DLCL already contains six latent classes, we asked whether the critique also applies to it. We fitted ML6 and the DLCL with L-BFGS from several starting points, with a small ridge (10^{-3} on the class offsets and context entries) to stop tiny classes from drifting to infinite coefficients, which otherwise happens with one choice per person. Our fits are much better than those in the paper: ML6 reaches an NLL of 9762.4 (paper: 9793) and the DLCL 9703.9 (paper: 9764 with Adam, 9709 with EM). With these fits the DLCL is a highly significant improvement over ML6 (LRT 117, 36 df, $p = 2 \times 10^{-10}$), unlike the paper's $p = 0.0089$.

That test covers all 36 context entries. For the oiliness effect we used two one-degree-of-freedom tests against ML6: the DLCL entry $A_{\text{oil,oil}}$, which acts only inside the oiliness component, and an LCL self-effect shared by all six classes (Table VII). With one choice per person, the shared effect is -0.21 ($p = 0.005$), below the paper's significance bar, while the DLCL entry is estimated at -7.3, a value that only makes sense inside one small component. With two or three ranked choices per person, as in the panel analysis, both oiliness tests are far from significant. The 36-entry DLCL test stays significant at $k = 1$ and $k = 3$.

TABLE VII
DLCL AND ML6 ON THE SUSHI DATA WITH THE TOP k CHOICES PER PERSON (p-VALUES; $A_{\text{oil,oil}}$ ESTIMATES IN PARENTHESES)

k	DLCL vs ML6	DLCL entry	ML6 + shared $A_{\text{oil,oil}}$
1	2×10^{-10}	4×10^{-4} (-7.3)	0.005 (-0.21)
2	0.008	0.79 (-0.04)	0.16 (-0.08)
3	1×10^{-9}	0.99 (-0.00)	0.84 ($+0.01$)

TABLE VIII
SIMULATION WITH DLCL AND ML6 (SHARE OF DATASETS WITH $p < 0.001$; MEAN $A_{\text{oil,oil}}$ IN PARENTHESES)

Ground truth	DLCL vs ML6	DLCL entry	ML6 + shared $A_{\text{oil,oil}}$
no effect	1%	0%	0% ($+0.04$)
true effect -0.27	8%	31%	82% (-0.26)
normal, $s = 0.75$	5%	1%	0% ($+0.01$)
normal, $s = 1$	9%	1%	0% ($+0.00$)
50/50 segments, $s = 1$	3%	1%	0% (-0.01)
20/80 segments, $s = 1$	0%	1%	0% (-0.02)

We repeated the simulation study with these tests (100 replicates per setting, run on Kaggle; Table VIII). Two points stand out. First, the DLCL does not share LCL's problem: with taste heterogeneity and no context effect, its oiliness tests flag at most 1% of datasets. The 36-entry test is somewhat too liberal (up to 9% at $p < 0.001$), but no heterogeneity-only dataset came close to the LRT of 117 seen on sushi (largest 91). Second, the DLCL is a poor tool for detecting an LCL-type self-effect. Against a true $A_{\text{oil,oil}} = -0.27$ its own entry detects the effect in 31% of datasets, with very unstable estimates (mean -2.6), and the 36-entry test in 8%, because each context entry in the DLCL acts only inside one component. A self-effect shared by all classes of ML6 detects it in 82% of datasets with no false positives, close to the random-coefficient test.

So the critique applies to LCL and to the interpretations built on it, while the DLCL's comparison with ML6 is a heterogeneity-aware test of the right kind. On sushi that test finds context structure that oiliness heterogeneity does not explain, but neither it nor ML6 supports the oiliness effect once each person's taste is pinned down by repeated choices.

I. Observation

Putting the pieces together, the sushi evidence weakens at each step. With one choice per person, the paper's test gives $p = 1.6 \times 10^{-6}$. Allowing tastes to vary moves this to about 10^{-3}, and the bootstrap says that the heterogeneity we can measure accounts for about a third of the estimate. Once each person's taste is pinned down by their second and third choices, the effect is gone at the later picks and only a smaller first-pick effect is left, at $p \approx 0.02$. The simulations make clear why the first step is misleading: a significant negative self-effect is exactly what an LCL reports for a population whose tastes differ, even when the menus are random.

The regularity argument limits the scope of this result. Heterogeneity can only produce effects that are consistent with a random utility model, so effects that imply regularity violations, or positive self-effects such as the paper's "truck on truck" effect in car-alt, need another explanation; for car-alt the authors found that menus were tailored to respondents. We also do not claim that sushi has no context effects: the test of all 36 entries remains significant after heterogeneity is modelled, both inside a random-coefficient model and in the DLCL's comparison with ML6.

Several limitations apply. The rank-ordered model assumes the same utility at every rank, while lower ranks are known to be noisier [16]; with $k = 5$ the panel estimate becomes $+0.12$ ($p = 9 \times 10^{-4}$), so we use $k \leq 3$, which we checked by simulation. Our heterogeneity models are diagonal normal or have at most three classes, and correlated tastes were not tried. The simulations use 100 replicates per setting (50 in the panel check), so rates are uncertain by about ± 5–10 points. The expansion is accurate only for small s.

IV. CONCLUSION

We reproduced the sushi results of the linear context logit and tested whether its oiliness self-effect can be explained by differences in taste between people. Theory and simulations on the real menus show that taste heterogeneity, with no context effect, produces a negative self-effect of the paper's size and significance, and that testing the effect inside a random-coefficient or latent-class model keeps the false-positive rate at the nominal level with little loss of power. On the real data, the fitted heterogeneity explains about a third of the effect with one choice per person. With two or three ranked choices per person, the effect is not distinguishable from zero at later picks, and a first-pick effect of about -0.15 remains at $p \approx 0.02$. The paper's DLCL, which already contains latent classes, does not share LCL's problem, but it is also not able to support the oiliness effect, and its individual entries are hard to estimate. We conclude that the paper's causal reading of the oiliness effect is not supported once taste heterogeneity is modelled. The practical lesson is simple: an LCL effect should be tested inside a heterogeneity model, using repeated choices per

person when the data have them. This keeps what we like about LCL, its interpretable and convex structure, and each of our fits took only seconds. Future work is to apply the same tests to the paper's other datasets and to try richer heterogeneity models for the remaining first-pick effect.

APPENDIX A
SOURCES OF CODE AND DATA

- Data: sushi preference data `sushi3-2016.zip` [2] from the author's website (https://www.kamishima.net/sushi/), kept in our private repository.
- Reference code: the feature preprocessing follows the authors' repository (https://github.com/tomlinsonk/feature-context-effects).
- Our code: `lcl_heterogeneity/src` (data loading, models, reproduction, expansion check, simulation study, real-data analysis, bootstrap, panel check and figures), with all outputs in `lcl_heterogeneity/results`.
- Libraries: PyTorch, NumPy, SciPy, pandas, Matplotlib.

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